

ENMT301 - Sensors & Signals Tutorial 2 solutions

$$1) I(s) = \frac{5}{s^2} - e^{-5s} \frac{5}{s^2} + \frac{5}{s^2 + 2.5 \times 10^5}$$

Table Pair 2
K=5, n=1

Table Pair 10
a=5

Table Pair 2
K=5, n=1

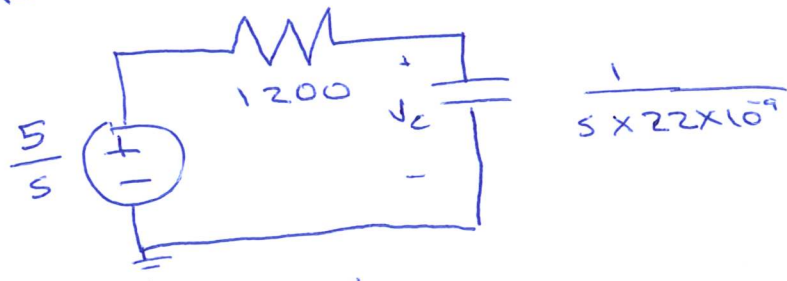
Table Pair 4
 $\omega = \sqrt{2.5 \times 10^5}$
 $\omega = 500$
K $\omega = 5$
K=0.01

$$i(t) = u(t) 5t - u(t-5) 5(t-5) + u(t) 0.01 \sin(500t)$$

2) a) Voltage source $V(t) = 5.0 u(t)$
 $V(s) = \frac{5.0}{s}$

Capacitor impedance = $\frac{1}{sC} = \frac{1}{s \times 22 \times 10^{-9}}$

Resistor impedance = $R = 1200$



$$V_c(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} \cdot V(s)$$

$$V_c(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} \cdot \frac{5}{s}$$

$$V_c(s) = \frac{\frac{5}{sC}}{Rs + \frac{1}{C}}$$

$$V_c(s) = \frac{5}{s^2 RC + s}$$

$$V_c(s) = \frac{5}{s(sRC + 1)}$$

Voltage across the capacitor

Using Table Pair 5 $K = 5$ $\tau = RC$

$$v_c(t) = K(1 - e^{-t/\tau}) u(t) \quad V$$

$$v_c(t) = 5(1 - e^{-3.8 \times 10^4 t}) u(t) \quad V$$

Current signal:

$$I(s) = \frac{V(s)}{Z(s)}$$

$$I(s) = \frac{\frac{5}{s}}{R + \frac{1}{sC}}$$

$$I(s) = \frac{5}{sR + \frac{1}{C}} \times \frac{\frac{1}{R}}{\frac{1}{R}}$$

$$I(s) = \frac{\frac{5}{R}}{s + \frac{1}{RC}}$$

Table Pair 3 $K = \frac{5}{R}$ $a = \frac{1}{RC}$

$$i(t) = K e^{-at} u(t)$$

$$i(t) = \frac{5}{R} e^{-t/RC} u(t)$$

$$i(t) = 4.2 \times 10^{-3} e^{-3.8 \times 10^4 t} u(t) \quad A$$

2)b) $v(t) = 6.0t u(t)$

$$V(s) = \frac{6.0}{s^2}$$

Table Pair 2
 $n=1, K=6$

Inductive impedance = sL

$$I_L(s) = \frac{V(s)}{Z(s)} = \frac{\frac{6.0}{s^2}}{sL}$$

$$= \frac{6}{L s^3} = \frac{6}{2L} \cdot \frac{2}{s^3}$$

Current through inductor

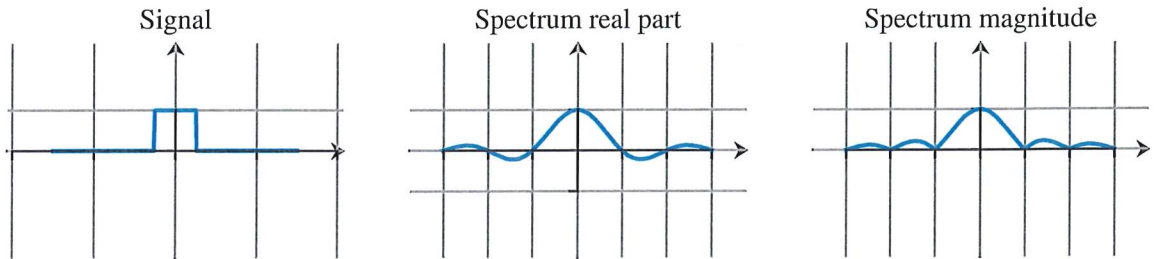
Table Pair 2 $K = \frac{6}{2L}, n=2$

$$i_L(t) = \frac{6}{2L} t^2 u(t)$$

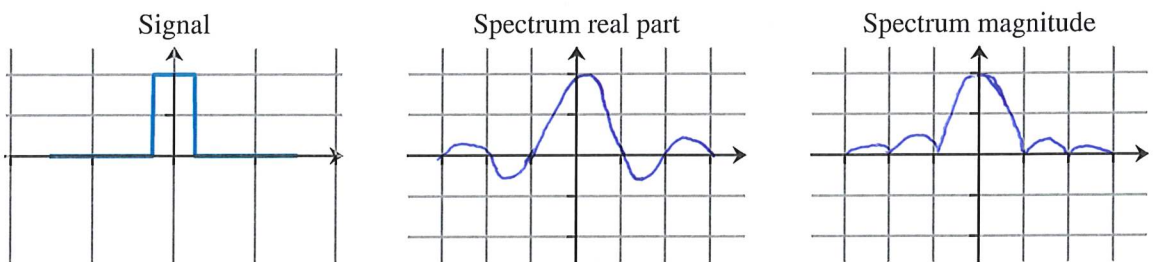
$$i_L(t) = 770 t^2 u(t) \quad A$$

Q3. Fourier Transforms

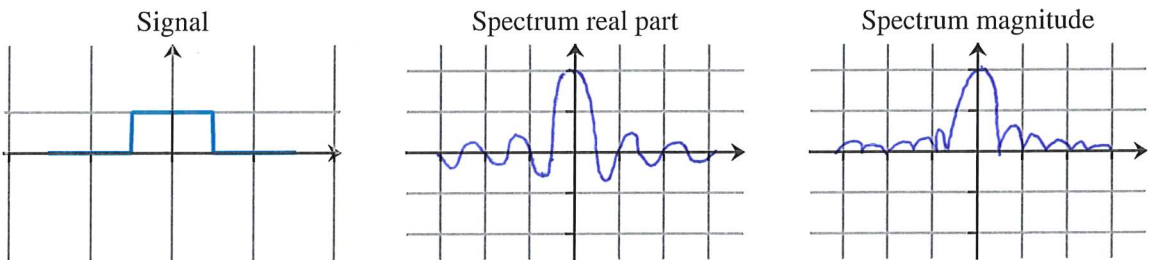
Complete the following Fourier transform pairs:



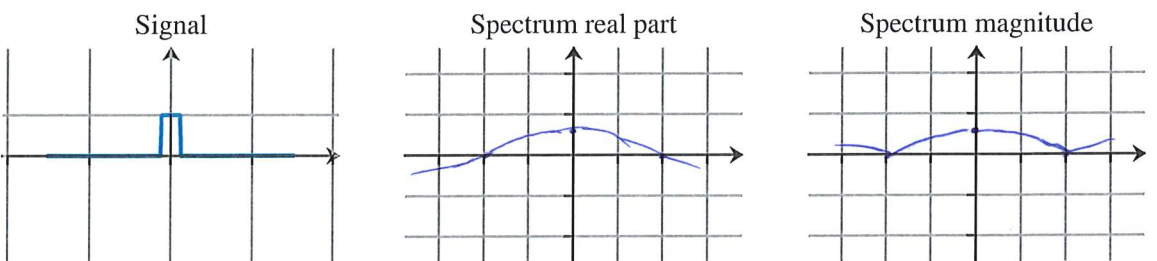
Signal double,
Spectrum
Doubled



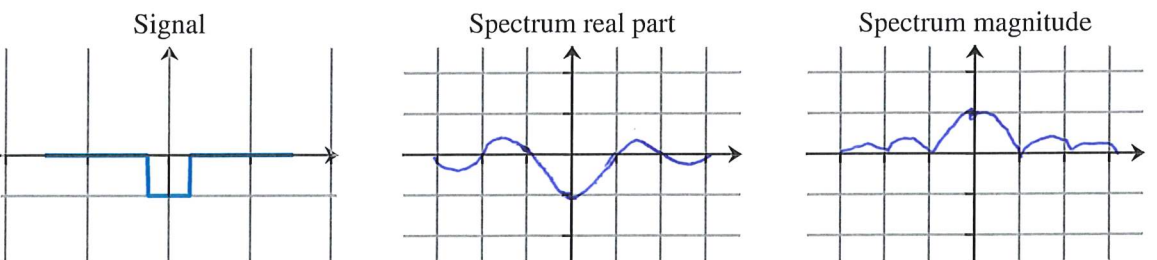
Signal twice
as wide,
Spectrum half
as wide and
doubled



Signal half
as wide,
Spectrum twice
as wide and
halved

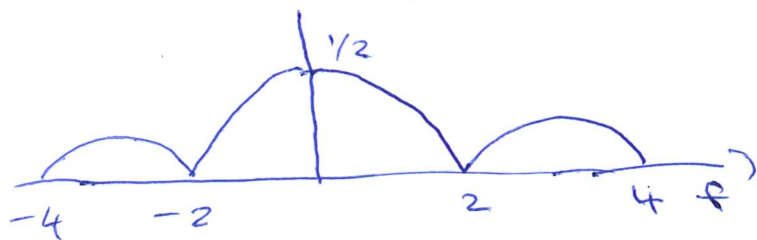
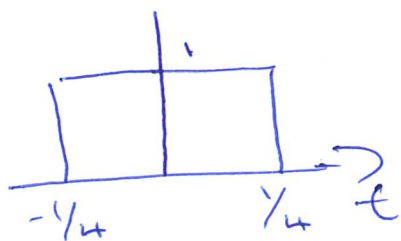


negative signal,
Spectrum
negated

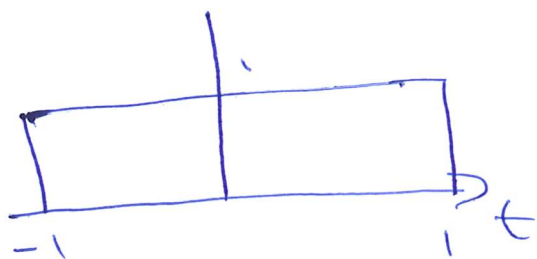


2) a) $\text{rect}(t) \longleftrightarrow \text{sinc}(f)$
 Use scaling theorem with $a = 2$

$$\text{rect}(2t) \longleftrightarrow \frac{1}{2} \text{sinc}\left(\frac{f}{2}\right)$$

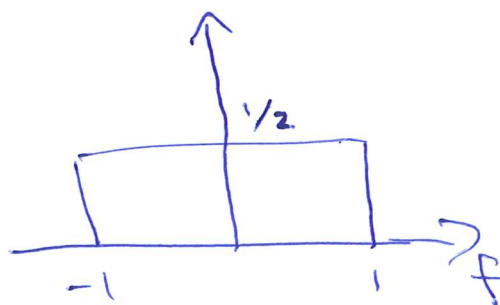
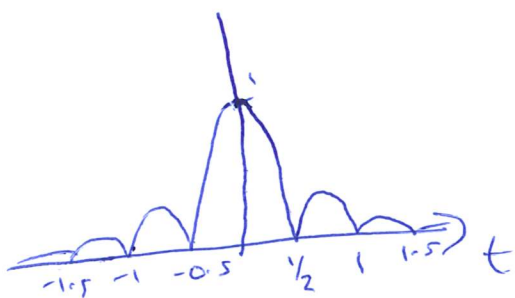


b) $\text{rect}(t/2) \longleftrightarrow 2 \text{sinc}(2f)$
 Scaling theorem with $a = 1/2$

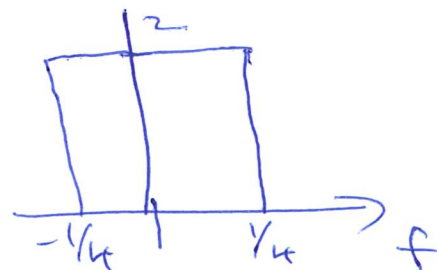
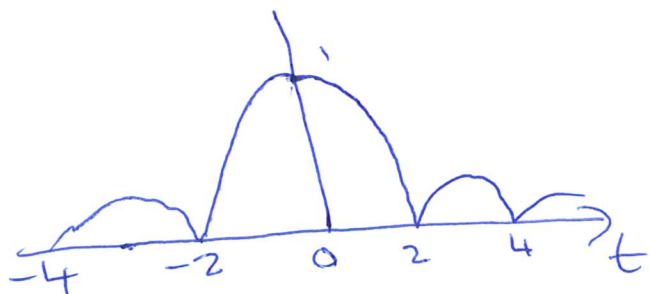


c) $\text{sinc}(t) \longleftrightarrow \text{rect}(f)$
 Use scaling theorem with $a = 2$

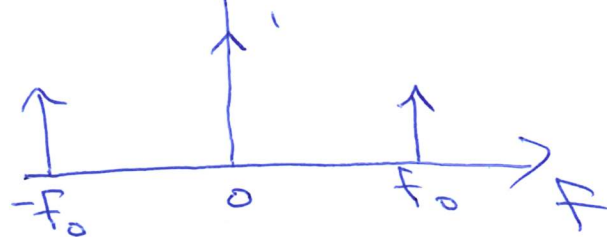
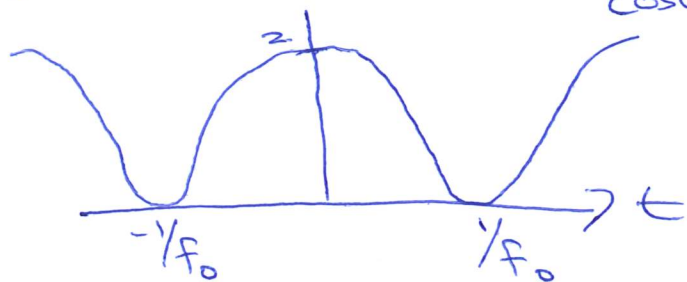
$$\text{sinc}(2t) \longleftrightarrow \frac{1}{2} \text{rect}\left(\frac{f}{2}\right)$$



d) $\text{sinc}(t/2) \longleftrightarrow 2 \text{rect}(2f)$
 Scaling theorem with $a = 1/2$

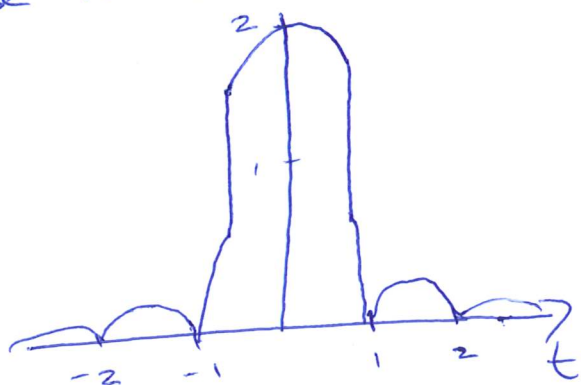


e) $1 + \cos(2\pi f_0 t) \longleftrightarrow \delta(f) + \frac{1}{2} \delta(f-f_0) + \frac{1}{2} \delta(f+f_0)$
 Use linearity theorem and $1 \longleftrightarrow \delta(f)$
 $\cos(2\pi f_0 t) \longleftrightarrow \frac{1}{2} \delta(f-f_0) + \frac{1}{2} \delta(f+f_0)$

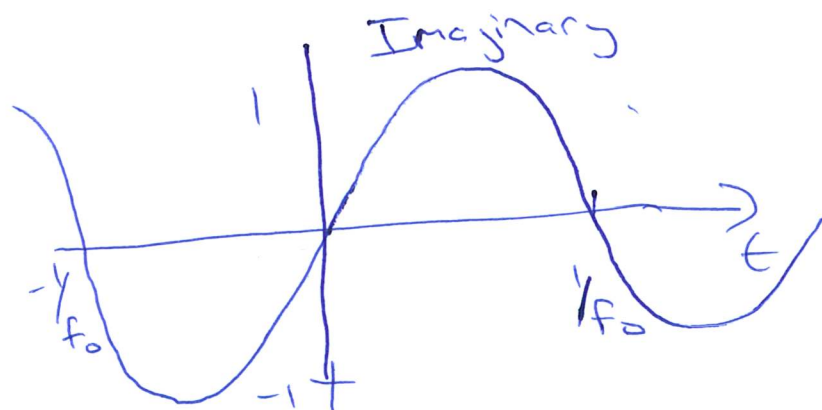
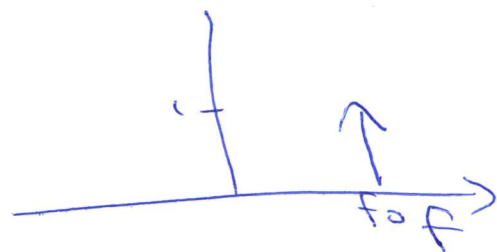
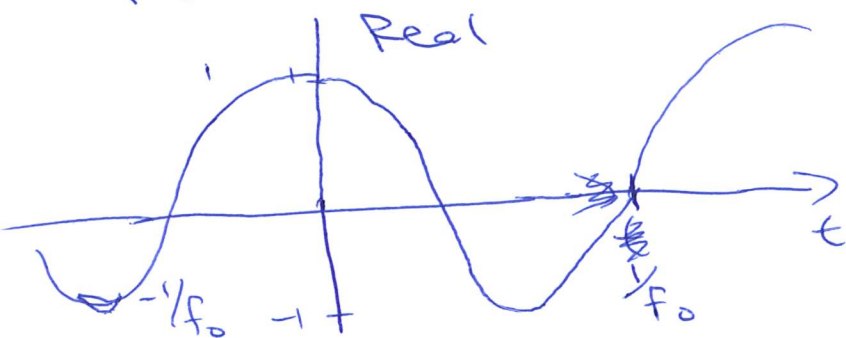


f) $\text{rect}(t) + \text{sinc}(t) \longleftrightarrow \text{sinc}(f) + \text{rect}(f)$
 Use linearity theorem and $\text{rect}(t) \longleftrightarrow \text{sinc}(f)$
 $\text{sinc}(t) \longleftrightarrow \text{rect}(f)$

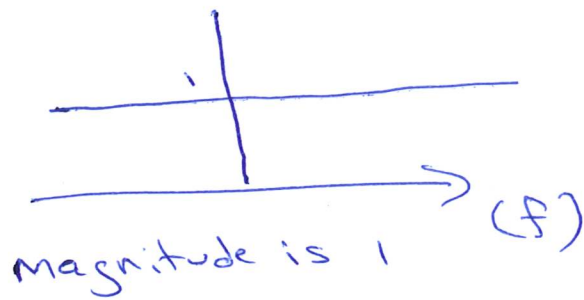
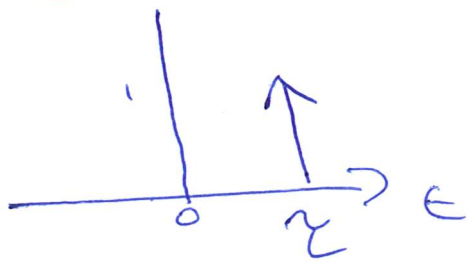
Same in
time & frequency
domains



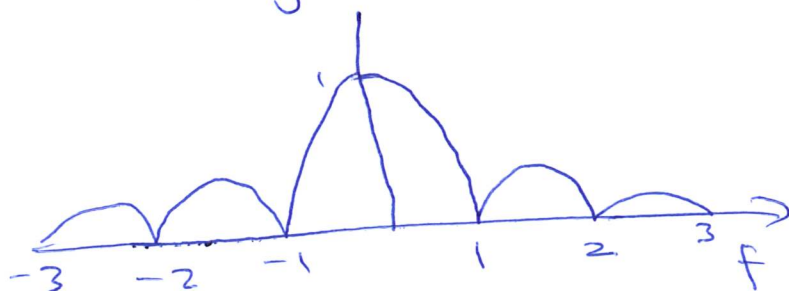
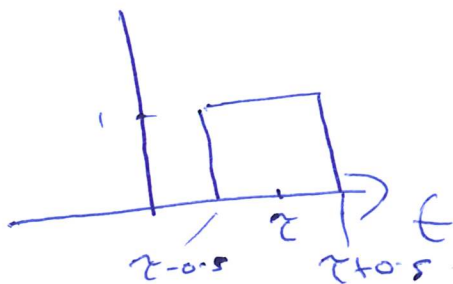
g) $\exp(j2\pi f_0 t) \longleftrightarrow \delta(f-f_0)$
 using frequency shift theorem and $1 \longleftrightarrow \delta(f)$
 $\exp(j2\pi f_0 t) = \cos(2\pi f_0 t) + j \sin(2\pi f_0 t)$



h) $\delta(t - \tau) \longleftrightarrow \exp(-j2\pi f\tau)$
 using shift theorem and $\delta(t) \longleftrightarrow 1$



i) $\text{rect}(t - \tau) \longleftrightarrow \text{sinc}(f) \exp(-j2\pi f\tau)$
 magnitude is sinc



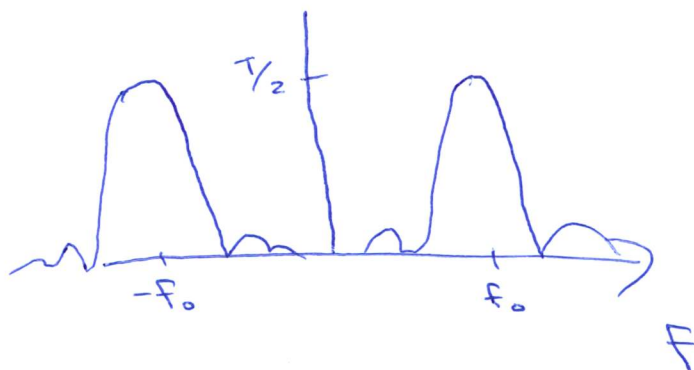
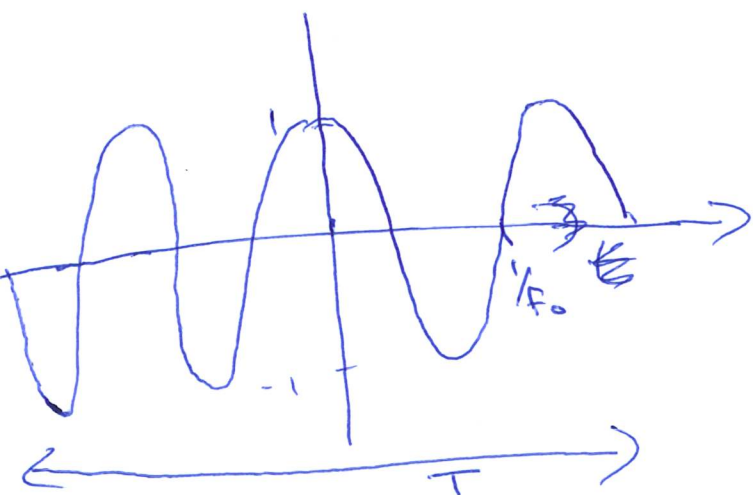
j) $\text{rect}\left(\frac{t}{T}\right) \cos(2\pi f_0 t)$

$\text{rect}\left(\frac{t}{T}\right) \longleftrightarrow T \text{sinc}(fT)$

$\cos(2\pi f_0 t) \longleftrightarrow \frac{1}{2} \delta(f + f_0) + \frac{1}{2} \delta(f - f_0)$

Use convolution theorem

$\text{rect}\left(\frac{t}{T}\right) \cos(2\pi f_0 t) \longleftrightarrow \frac{T}{2} \text{sinc}((f + f_0)T) + \frac{T}{2} \text{sinc}((f - f_0)T)$



$$\begin{aligned}
 \text{5) a) } \text{rect}(f) &\longleftrightarrow \text{sinc}(t) \\
 \text{rect}(2f) &\longleftrightarrow \frac{1}{2} \text{sinc}\left(\frac{t}{2}\right) \\
 \text{b) } \text{rect}\left(\frac{f}{2}\right) &\longleftrightarrow 2 \text{sinc}(2t) \\
 \text{c) } \text{sinc}(f) &\longleftrightarrow \text{rect}(t) \\
 \text{sinc}(2f) &\longleftrightarrow \frac{1}{2} \text{rect}\left(\frac{t}{2}\right) \\
 \text{d) } \text{sinc}\left(\frac{f}{2}\right) &\longleftrightarrow 2 \text{rect}(2t)
 \end{aligned}$$

$$\begin{aligned}
 \text{e) } \delta(f) &\longleftrightarrow 1 \\
 \delta(f-2) + \delta(f+2) &\longleftrightarrow e^{j4\pi t} + e^{-j4\pi t} = 2 \cos(4\pi t) \\
 \text{f) } \delta(f-2) - \delta(f+2) &\longleftrightarrow e^{j4\pi t} - e^{-j4\pi t} = 2j \sin(4\pi t) \\
 \text{g) } \delta(f-2) &\longleftrightarrow e^{j4\pi t}
 \end{aligned}$$